# Safety systems · Czechia

České dráhy

Czech main signals combine **this signal's speed** (the lower yellow lamp, with light strips below the head: none = 40, yellow strip = 60, one green = 80, two green = 100 km/h) and **what the next signal shows** (the upper lamps): green = volno, yellow = výstraha (stop), slow-flashing yellow = expect 40, fast-flashing yellow = expect 60, slow-flashing green = expect 80, fast-flashing green = expect 100. The **LS** cab system repeats the signals where the track carries codes; **Liberec – Stará Paka has no codes**, so there it only checks your vigilance. Speeds in km/h. Quadrant controls are from TheJAG's ČD 750 and 843 profiles.

## Safety systems

### LS (LS 90 · LS IV / LVZ)

Liniový vlakový zabezpečovač

On coded track the cab repeater shows the next signal: green, yellow, yellow ring (reduced speed), red; **blue = no vigilance press needed**. A more restrictive code asks for a press. On uncoded track (this route) it runs the vigilance check only.

**WHERE** 843: LS 90. 750: LS IV type LVZ (same behaviour)

**QUADRANT** 843: left aux = LS 90 (On / Isolated); 750: left aux = LVZ (Active / Off)

### Vigilance

Kontrola bdělosti

When the **blue lamp goes out**, press the vigilance button within about **7 s**; then a **siren** for 5 s more, then the emergency brake. The checks pause while the loco brake is applied.

**WHERE** ČD 750 and 843

**QUADRANT** **Gear lever** = vigilance button

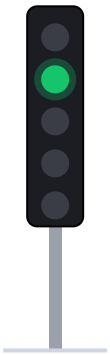
## Main signals (hlavní návěstidla)



### Stůj

Stop.

**DRIVE** Stop before the signal.



### Volno

Proceed; next signal at least proceed.

**DRIVE** Line speed.



### Výstraha

Next signal at Stůj.

**DRIVE** Brake to stop at the next signal.



### Očekávej 40

Slow-flashing yellow: next signal 40 km/h.

**DRIVE** Brake to 40 by the next signal.



### Rychlost 40 a volno

40 km/h from here (until the last points are cleared); next signal proceed.

**DRIVE** Be at 40 at the signal.



### Rychlost 40 a výstraha

40 km/h from here; next signal at Stůj.

**DRIVE** Be at 40, then brake to stop.



### Rychlost 60 a volno

Lower yellow + yellow strip: 60 km/h (one green strip 80, two 100).

**DRIVE** Be at 60 at the signal.



### Přivolávací návěst

Red + flashing white: pass the signal at Stůj.

**DRIVE** Max 40 on sight, ready to stop.

Sources: Vector Simulations: Liberec – Stará Paka manual · DTG: Liberec signalling deep dive · vyhláška 173/1995 Sb. příloha 1 / SŽ D1 · cs.wikipedia: Železniční návěstidlo, Liniový vlakový zabezpečovač LS



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## Distant signals, boards and shunting



### Předvěst: výstraha

Distant signal: the main signal shows Stůj (flashing = expect a speed).

**DRIVE** Brake to stop at the main signal.




### Předvěstník

A lower limit ahead: 60 km/h (tens).

**DRIVE** Brake to reach it by the rychlostník.




### Rychlostník

The limit from here, in km/h.

**DRIVE** Lower: by the board. Higher: once the rear has passed.




### Vzdálenostní upozorňovadlo

Countdown boards (3, 2, 1 stripes) to a distant signal.

**DRIVE** Watch for the distant signal.



### Posun dovolen

Shunting allowed.

**DRIVE** Shunt only: max 40 (30 propelling).

## How to drive under LS and the vigilance check

### On this route (no codes)

1. Drive to the lineside signals; the repeater shows no code.
2. When the **blue lamp goes out**, press **vigilance** (gear lever) within about 7 s.
3. Missed, the siren sounds: you have 5 s more, then the emergency brake.

### After an LS emergency brake

1. Stop.
2. 843: switch LS 90 off and on (left aux twice).
3. 750: switch the LVZ off and on at the console in the engine room (or the left aux), then release the brake.

### With LS codes (main lines)

- The repeater shows the next signal: green, yellow, yellow ring (reduced speed), red.
- Blue lit: no press needed.
- A more restrictive code: press **vigilance** at once, then brake to the signal.

### Before departure

1. Loco brake fully applied.
2. Train protection to **Start**, wait for the start lamp, then **On / Active** (843 LS 90, 750 LVZ). The profile's left aux sets On directly; if the system does not arm, use Start in the cab first.
3. Check the signal, close the doors, release the brake.

**Remember:** The lower yellow lamp and the strips below it are the speed at this signal; the upper lamp tells you what the next signal shows.

Sources: Vector Simulations: Liberec – Stará Paka manual · DTG: Liberec signalling deep dive · vyhláška 173/1995 Sb. příloha 1 / SŽ D1 · cs.wikipedia: Železniční návěstidlo, Liniový vlakový zabezpečovač LS



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